

Lucas/Delphi DP 200 Series fuel injection pump

General description

These pumps have a locking screw (A3) and a spacer (A2). The locking screw prevents the rotation of the drive shaft when the pump is removed from the engine.

An "O" ring (A1) is fitted in a groove in the pump flange. This "O" ring is fitted instead of a joint between the pump flange and the timing case.

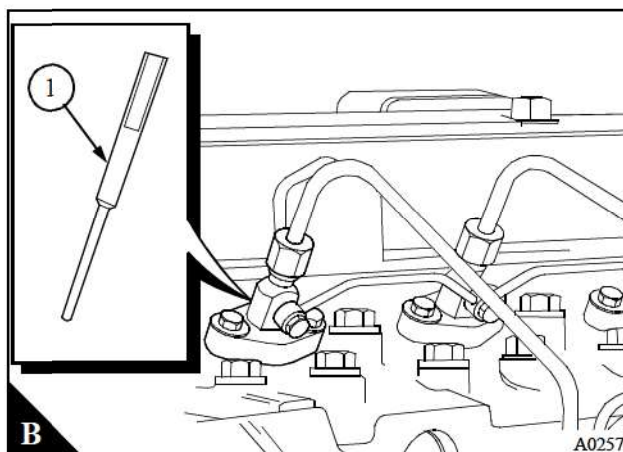
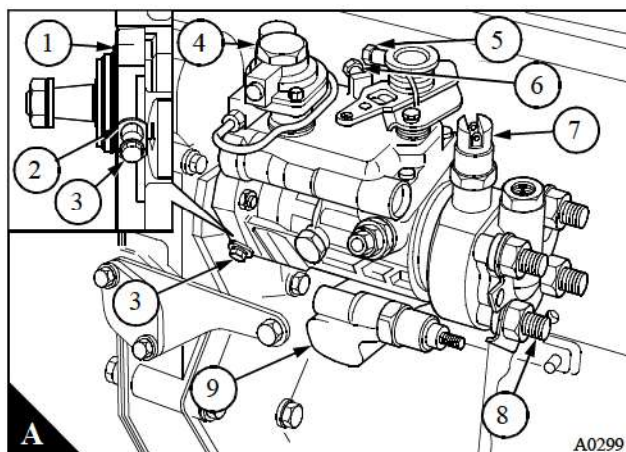
Caution: The drive shaft of the pump must not be rotated without the spacer in position under the locking screw. If the drive shaft is rotated, the drive shaft will be damaged.

Engines with these pumps have a timing mark on the pump flange but not on the timing case.

For four cylinder engines the high-pressure outlet for number 1 cylinder is shown at (A8).

A new pump will be received locked at the correct angle for the engine. It is important that the engine number is given to the distributor when a new pump is needed.

These engines operate with the timing of the fuel injection retarded, therefore the accuracy necessary to position the piston for timing is difficult to obtain. Greater accuracy is possible if the piston is further down the cylinder bore. A piston position probe (B1), tool 21825630 or 21825947, is used to set accurately the piston at 100° BTDC.



To remove

Operation 11-38

Special requirements

Special tools			
Description	Part number	Description	Part number
Piston position probe	21825630 or 21825947	Gear puller	21825565

1 Remove the rocker cover.

2 Loosen the setscrews which retain the atomisers.

3 Rotate the crankshaft clockwise, from the front, until the push rod for the inlet valve of number 1 cylinder just releases.

4 Remove the atomiser together with its seat washer from number 1 cylinder and fit the piston position probe, 21825630 or 21825947 (A1) instead. Fit the atomiser clamp to the probe and tighten the setscrews gradually and evenly to 12 Nm (9 lbf ft) 1,2 kgf m.

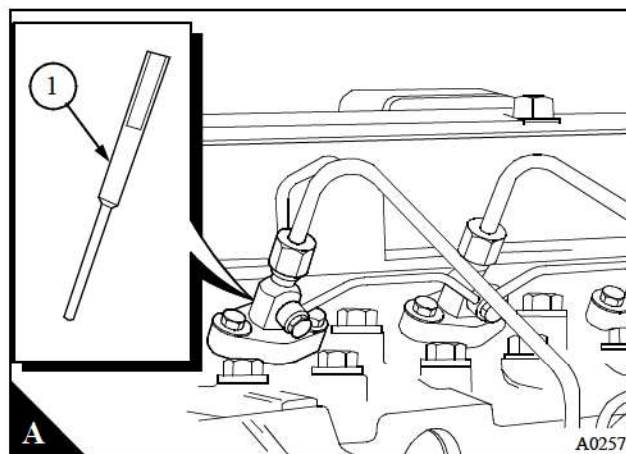
Caution: Ensure that the piston comes lightly into contact with the piston position probe, or both the piston and the probe could be damaged.

5 Slowly rotate the crankshaft clockwise, from the front, until the piston just comes into contact with the piston position probe. The piston will now be set at 100° BTDC on the compression stroke of number 1 cylinder.

Caution: Do not rotate the crankshaft when the pump is not on the engine; the loose fuel pump gear may damage the timing case. If it is necessary to rotate the crankshaft, fit the fuel pump temporarily to ensure that the gear is in the correct position. If the fuel pump is fitted temporarily in order to rotate the crankshaft, the locking screw must be released.

Notes:

- The fuel pump can be removed from the engine and fitted again if:
- The crankshaft is not rotated
- The drive shaft of the fuel pump is not rotated



Continued

Caution: If the crankshaft or the pump shaft is rotated when the pump is off the engine. Or if the engine timing is not correct the timing of the fuel injection pump must be checked, see Operation 8-14.

6 Release the locking screw and adjust the spacer (B1) to enable the locking screw (B2) to be tightened on the drive shaft of the fuel pump. Tighten the locking screw to 12 Nm (9 lbf ft) 1,2 kgf m. Check that the spacer is free to move. The drive shaft of the fuel pump is now fastened.

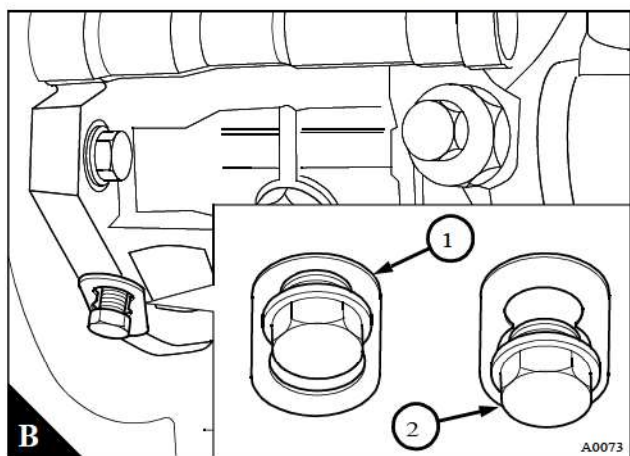
Caution: Use a second spanner to prevent movement of the high-pressure outlet when the union nut for each high-pressure pipe is released.

7 Remove the pipes, the cables and the connections for the cold start device and the electrical stop solenoid from the fuel pump. Loosen the nuts on the pump flange.

8 Release the setscrews and remove the gear cover from the timing case cover. Release the nut which retains the drive gear of the pump one to two turns.

9 Fit the gear puller 21825565 and loosen the gear on the drive shaft of the pump. Remove the puller, nut and spring washer.

10 Remove the nuts from the flange of the fuel pump and remove the pump.



To fit

Operation 11-39

- 1 Inspect the "O" ring and, if necessary, fit a new "O" ring.
 - 2 Lightly lubricate the "O" ring with clean engine lubricating oil and put the pump into position on the timing case. Ensure that the key is engaged correctly in the keyway of the drive gear.
 - 3 Fit the spring washer and the nut to retain the drive gear. Tighten the nut to approximately 15 Nm (11 lbf ft) 1,5 kgf m.
 - 4 Hold the top of the pump toward the engine to remove the backlash and fit the nuts to the pump flange. Tighten them to 22 Nm (16 lbf ft) 2,2 kgf m.
 - 5 Release the locking screw on the pump and adjust the spacer (A1) to prevent the locking screw (A2) from locking the pump shaft, tighten the locking screw to 12 Nm (9 lbf ft) 1,2 kgf m. Check that the spacer cannot move. The drive shaft of the fuel pump is now free to move.
 - 6 Fully tighten the nut for the drive gear to 80 Nm (59 lbf ft) 8,2 kgf m. Fit the gear cover.
- Caution:** Use a second spanner to prevent movement of the high-pressure outlet when the union nut for each high-pressure pipe is tightened.
- 7 Remove the piston position probe and fit the atomiser together with a new seat washer. Tighten the setscrews of all of the atomisers gradually and evenly to 12 Nm (9 lbf ft) 1,2 kgf m.
 - 8 Fit the pipes, cables and connection for the cold start device and electrical stop solenoid to the pump.
 - 9 Eliminate air from the fuel system, see Operation 11-46.
 - 10 Fit the cylinder head rocker cover, see Operation 3-2.
 - 11 Operate the engine and check for leakage. With the engine at the normal temperature of operation, check that the idle speed and the maximum no load speed are correct, see Operation 11-40.

